

Home Exam MVE441

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1 Question 1(a)

The dataset used was B11, containing 7680 training points with 820 input features across 7 classes. Label distribution is shown in table 1. Input values range from -12.725 to 14.072 , with a mean of 0.419 and standard deviation of 3.328 .

For dimensionality reduction, PCA was first applied but showed no clear elbow. Instead a $\frac{1}{x}$ -shaped variance curve, indicating high variance across all features without class-specific information. Since PCA is unsupervised, F-statistics (`sklearn.feature_selection.r_regression`) were used instead to identify class-correlated features. A clear elbow appeared at around 20 features (figure 1), which were adopted as the primary features for the remainder of the project. Since the F-test is linear, mutual information (`sklearn.feature_selection.mutual_info_regression`) was also applied to catch any nonlinear correlations, but it identified the same features. All selected features had p-values well below 0.05 . To prevent adding features with 1 to 1 correlation, a correlation matrix were created (table 2), where it's clear that no features have a correlation of 1 between each other.

Three models were selected: Logistic Regression (PyTorch), kNN (sklearn), and Regularized Discriminant Analysis (NumPy). Logistic Regression provides a strictly linear decision boundary, RDA models each class with Gaussian distributions and adapts well to varying data sizes, and kNN makes no distributional assumptions while capturing highly complex boundaries, three contrasting approaches.

Model evaluation followed the same nested cross-validation pipeline as project 3: the inner kCV ran 3 times, the outer 5 times, repeated 10 times to assess variance, where the kCV always used stratification to create folds. Hyperparameters were tuned using Bayesian optimisation (scikit-optimize), with 20 evaluations per outer fold where the 10 first had a focus on exploration and 10 last had a focus on performance. Models were evaluated on accuracy and macro F1-score, where each class contributes equally regardless of its size. This was done to have a score that showed if a model was down-prioritizing minority classes. Model comparison used Bayesian analysis, with the ROPE determined by running the pipeline on randomly labelled data and set such that the mean probability of equivalent performance exceeded 95%, and this was set to 4%.

The results of the pipeline can be found in figure 2. We can see that the performance is around high 80% to low 90% for the 200 sample size, and goes rapidly up to around 95% for 1000 sample sizes and from there slowly creeps upward from this. We also see that the variance of the scores are slowly reduces with the increase of sample size which is expected. Both scores is quite similar with the exception that f1-score is a bit lower for Logistic Regression and kNN in the beginning. This makes sense if we look at the class accuracy in figure 3. We can see that for 200 samples both especially kNN had a hard time classifying class 3 which makes sense as this is by far the smallest class in the dataset. If we look at the variance of the accuracy of class 3 we can also see that this is the largest for all the models when the sample size is small, which is expected. Looking at figure 4 we can see that in the beginning the Regularized Discriminant

Analysis is the clear winner but around 1000 samples they all score within the rope. This is expected because of RDAs ability to adapt it self after small sample sizes.

1.1 Conclusions

- The dataset is fairly balanced where most of the classes has appears around 15% with the exception of class number 3 that only has 6.2% of the labels.
- The index of features that were relevant for the classification task where the following: 7, 37, 100, 141, 158, 179, 210, 228, 276, 308, 410, 427, 519, 550, 558, 597, 752, 762, 796, 809. Checking the correlation matrix for these matrices confirmed that none of these features were exactly the same.
- The models does a good job of classifying the dataset with accuracy and F1 scores around 90% for most of the sample sizes. KNN and Logistic Regression does a bit worse on the smallest sample sizes compared to RDA but this expected because of RDAs ability to adapt with LDA and Native Bayes which is known to performe great on smaller sample sizes.
- For smaller sample sizes the accuracy and performance for class 3 is noticable worse than for other classes which is expected. For larger sample sizes the hardest class instead becomes class 7 for all the models. This could be because of the class distribution in features.

Class	1	2	3	4	5	6	7
Fraction	15.6%	15.6%	6.2%	15.8%	14.1%	15.6%	17.2%

Table 1: This table shows the balance over the dataset. The smallest class of the dataset is class 3 and the largest class is 7.

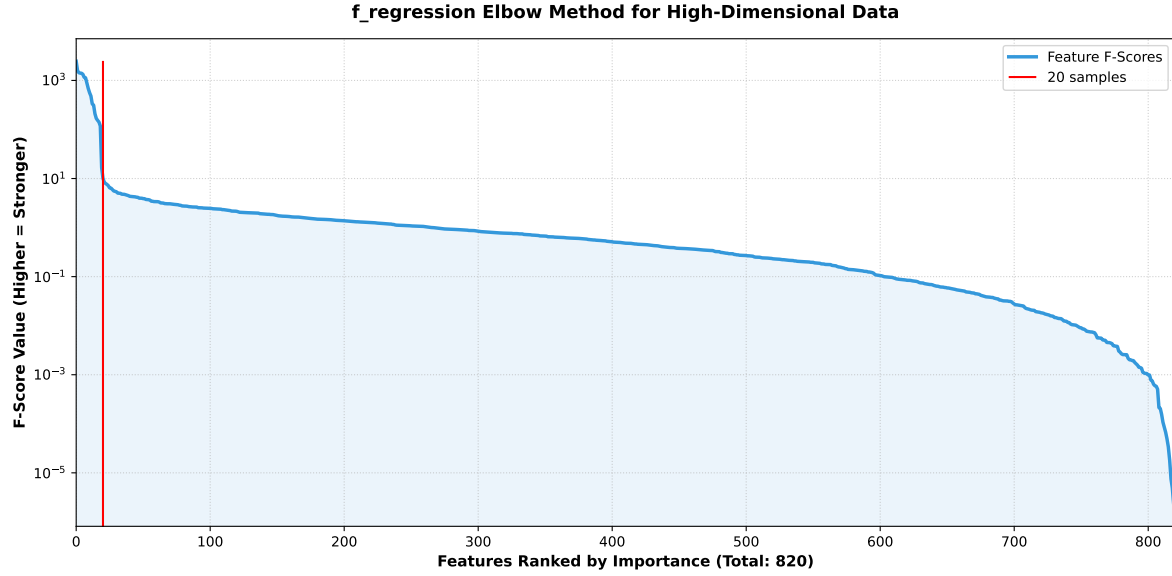


Figure 1: In this figure we can see the F-Score plotted from highest to lowest. We can see a clear elbow after 20 features, and therefore these are the features we will use in the rest of the project.

	7	37	100	141	158	179	210	228	276	308	410	427	519	550	558	597	752	762	796	809
7	1.00	0.18	0.49	0.33	-0.40	0.59	-0.39	0.24	-0.27	0.51	-0.19	-0.06	-0.08	0.33	-0.33	-0.35	0.18	0.69	-0.21	-0.05
37	0.18	1.00	0.06	-0.12	0.41	0.19	0.23	0.32	0.54	0.37	-0.50	-0.35	-0.30	-0.19	0.48	-0.18	-0.27	0.18	-0.08	-0.21
100	0.49	0.06	1.00	0.16	-0.37	0.51	-0.52	0.24	-0.15	0.54	-0.39	-0.10	-0.16	0.51	-0.30	-0.47	-0.05	0.60	-0.18	-0.19
141	0.33	-0.12	0.16	1.00	-0.39	0.41	-0.24	-0.01	-0.31	0.31	0.19	0.23	0.24	0.31	-0.28	-0.13	0.28	0.46	0.24	0.03
158	-0.40	0.41	-0.37	-0.39	1.00	-0.30	0.50	0.07	0.72	-0.18	-0.28	-0.31	-0.15	-0.37	0.69	0.17	-0.46	-0.45	0.17	-0.22
179	0.59	0.19	0.51	0.41	-0.30	1.00	-0.40	0.23	-0.07	0.56	-0.37	-0.14	-0.04	0.31	-0.21	-0.50	0.10	0.69	-0.03	-0.20
210	-0.39	0.23	-0.52	-0.24	0.50	-0.40	1.00	0.02	0.36	-0.20	0.16	0.07	-0.06	-0.40	0.51	0.41	-0.22	-0.45	0.01	0.13
228	0.24	0.32	0.24	-0.01	0.07	0.23	0.02	1.00	0.23	0.47	-0.34	-0.22	-0.24	-0.06	0.14	-0.17	-0.18	0.24	-0.14	-0.03
276	-0.27	0.54	-0.15	-0.31	0.72	-0.07	0.36	0.23	1.00	0.12	-0.53	-0.38	-0.27	-0.36	0.69	-0.05	-0.51	-0.21	0.13	-0.31
308	0.51	0.37	0.54	0.31	-0.18	0.56	-0.20	0.47	0.12	1.00	-0.45	-0.16	-0.26	0.19	-0.05	-0.44	-0.10	0.67	-0.18	-0.11
410	-0.19	-0.50	-0.39	0.19	-0.28	-0.37	0.16	-0.34	-0.53	-0.45	1.00	0.52	0.35	-0.01	-0.25	0.52	0.37	-0.32	0.07	0.46
427	-0.06	-0.35	-0.10	0.23	-0.31	-0.14	0.07	-0.22	-0.38	-0.16	0.52	1.00	0.29	0.18	-0.27	0.29	0.30	-0.10	0.17	0.38
519	-0.08	-0.30	-0.16	0.24	-0.15	-0.04	-0.06	-0.24	-0.27	-0.26	0.35	0.29	1.00	0.19	-0.14	0.16	0.22	-0.09	0.26	0.06
550	0.33	-0.19	0.51	0.31	-0.37	0.31	-0.40	-0.06	-0.36	0.19	-0.01	0.18	0.19	1.00	-0.38	-0.23	0.14	0.37	0.08	-0.16
558	-0.33	0.48	-0.30	-0.28	0.69	-0.21	0.51	0.14	0.69	-0.05	-0.25	-0.27	-0.14	-0.38	1.00	0.15	-0.41	-0.34	0.11	-0.16
597	-0.35	-0.18	-0.47	-0.13	0.17	-0.50	0.41	-0.17	-0.05	-0.44	0.52	0.29	0.16	-0.23	0.15	1.00	-0.04	-0.54	0.04	0.34
752	0.18	-0.27	-0.05	0.28	-0.46	0.10	-0.22	-0.18	-0.51	-0.10	0.37	0.30	0.22	0.14	-0.41	-0.04	1.00	0.19	0.02	0.23
762	0.69	0.18	0.60	0.46	-0.45	0.69	-0.45	0.24	-0.21	0.67	-0.32	-0.10	-0.09	0.37	-0.34	-0.54	0.19	1.00	-0.22	-0.14
796	-0.21	-0.08	-0.18	0.24	0.17	-0.03	0.01	-0.14	0.13	-0.18	0.07	0.17	0.26	0.08	0.11	0.04	0.02	-0.22	1.00	-0.10
809	-0.05	-0.21	-0.19	0.03	-0.22	-0.20	0.13	-0.03	-0.31	-0.11	0.46	0.38	0.06	-0.16	-0.16	0.34	0.23	-0.14	-0.10	1.00

Table 2: A table over the correlation matrix for the 20 features that was picked out from the F-Test. No other correlation is close to 1.00 in this table which shows that all of these features are needed for the classification task.

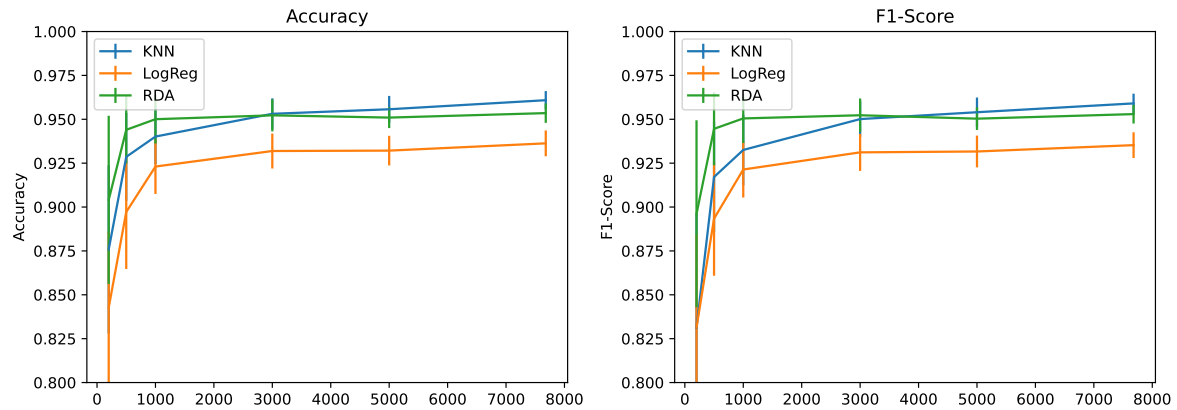


Figure 2: In this figure presents the performance for the different models after running the pipeline over different sample sizes. The left plot displays the accuracy score and the right plot shows the F1-score. We can see that Logistic Regression is performing the worst, and the KNN and RDA is almost similar in their performance over different sample sizes.

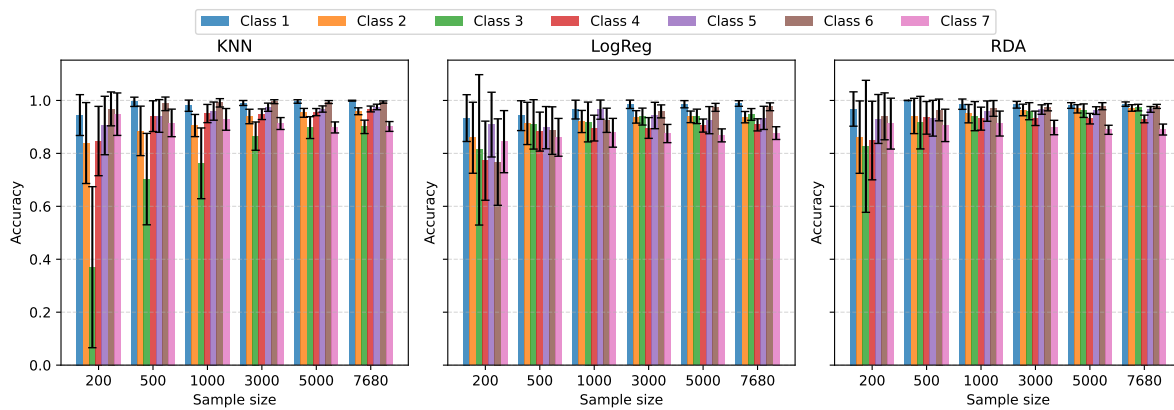


Figure 3: In this figure we can see each model's accuracy per class. While performing the worst for smaller sample sizes, especially for class 3, the accuracy is quite similar for higher sample sizes.

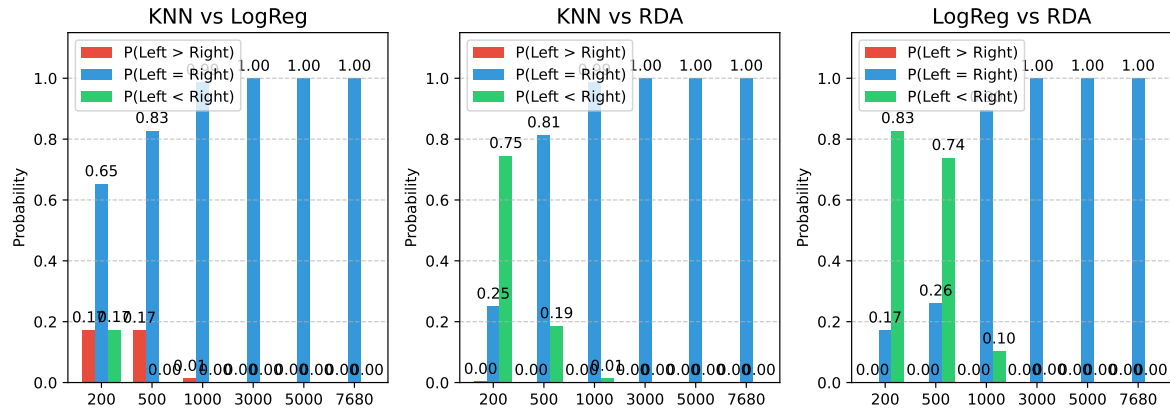


Figure 4: In this picture we can see the bayesian comparisons between the models, where the performance score used is F1-score. While RDA is performing the best in the beginning, the KNN and LogReg is fast to catch up around 1000 in sample size. The rope that was used for the comparison were 4%.

2 Question 1(b)

For this question.

2.1 Conclusions

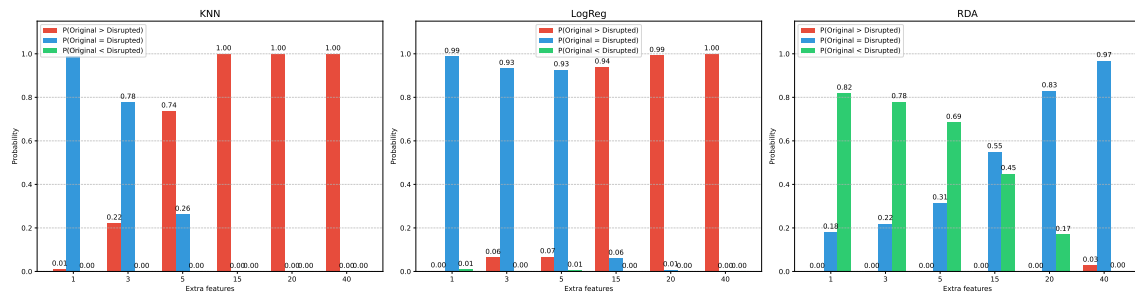


Figure 5: